

Submitting AI Jobs to the Anvil Supercomputer

Beginner Track 1 — AI for All Workshop, Sept 15 2026, The Mill at MSU Starkville

90-minute hands-on session (1:00–2:30 PM CT). Attendees learn to submit and monitor AI jobs on the Anvil supercomputer (Purdue/ACCESS/NAIRR) using Slurm.

Contents

File	What it is
lesson-plan.md	Primary deliverable: minute-by-minute 90-min plan
slides-outline.md	Slide-by-slide outline (~18 slides)
handout.md	Attendee lab guide (follows the 4 labs)
instructor-notes.md	Pre-session checklist, timing cues, pitfalls, fallbacks
labs/hello.py + labs/hello.sbatch	Lab 1: first job, inspect the Slurm environment
labs/train_digits.py + labs/train_digits-cpu.sbatch	Lab 2: real AI job (scikit-learn digits, CPU, no download)
labs/train-gpu.py + labs/train-gpu.sbatch	Lab 3: PyTorch GPU job (gpu-debug partition)
labs/array.sbatch	Lab 4 (stretch): job arrays

Ground truth (verified from RCAC docs, Sept 2026)

- Anvil: 1,000 nodes, 2x 64-core AMD EPYC Milan per node (128 cores), Slurm scheduler, 16 nodes x 4x NVIDIA A100 (partition `gpu`), 21 nodes x 4x H100 (partition `ai`).
- Login: `ssh <user>@anvil.rcac.purdue.edu` (SSH keys only; username is `x-<ACCESSname>`) or Open OnDemand: <https://ondemand.anvil.rcac.purdue.edu>
- Mandatory sbatch fields: `#SBATCH -A <account>` (find via `mybalance`) and `#SBATCH -p <partition>` (default is `shared` if omitted).
- Partitions: `debug` (2 hrs, quick tests), `gpu-debug` (30 min, 2 GPUs), `shared` (default, 128 cores, 96 hrs), `wholenode`, `wide`, `highmem`, `gpu`, `ai`.
- Software: Lmod modules (`module load conda -> Anaconda 2021.05`), Singularity, NGC containers on GPU nodes.
- Docs: <https://docs.rcac.purdue.edu/userguides/anvil/> (Job Submission: `.../jobs/`)

Adaptation decisions still open (see instructor-notes.md)

1. Which account string attendees use (ACCESS credit transfer vs. NAIRR allocation vs. a workshop-provided allocation) — confirm with Preston Smith / RCAC before the session.
2. Exact GPU PyTorch module or NGC container for Lab 3 — run `module spider pytorch` on Anvil login node during pre-session prep.